

1] Process Management in linux

Linux is a multi-user and multi-tasking operating system. This means that many users and many programs can run simultaneously. Each running program is treated as a process and is assigned a unique number called process ID (PID).

Processes consume system resources such as:

- ① CPU time
- ② Main memory (RAM)
- ③ Disk and I/O resource

When unnecessary or faulty processes continue running they increase system load and reduce performance. To manage this linux provides command like:

- ① Ps
- ② top
- ③ kill
- ④ killall
- ⑤ Pkill

2] Identifying running processes

Before killing any process, Admin must first identify which processes are running and which ones are consuming more system resources.

This can be done using the following commands

a] Ps command

The Ps (process status) command displays information about active processes.

Ex- Ps -ef

This command shows:

PID, user who started the process, process name, CPU usage.

b] top command

The top command displays processes in real-time and shows.

CPU usage, memory usage, running processes

By using Ps or top, admin can decide which process should be terminated to reduce system load.

c] killing processes by name

a] killall command

The killall command is used to terminate all processes having the same name.

Syntax- killall process-name

Theory- killall sends a terminate signal to all processes with the specified name. It is useful when multiple copies of the same program are running. It reduces system load quickly by stopping all related processes at once.

Ex- killall chrome

4] killing a process based on process name
Linux also provides the pkill command to kill processes using their name or pattern.

Syntax- `pkill process_name`

Theory- pkill terminates process by matching their names. It is more flexible than killall because it can match partial names or patterns. It allows users to stop process without knowing their PID.

Ex- `pkill firefox`

5] killing a single process using process PID
Sometimes adam wants to terminate only one specific process without affecting others. For this purpose, linux uses the kill command along with the process ID.

Step 1 - Find the PID

Adam first finds the PID of the process using:

`Ps -ef | grep process_name`

Step 2 - kill the process using PID
`kill PID`

Theory- The kill command sends a termination signal to the process.

Three Radhe The process is given a chance to close properly.

Ex- kill 2456

6] Force killing a process

Sometimes a process not respond to normal kill commands. It may be stuck or frozen. In such cases, Admin uses a force kill command.

Syntax kill -9 PID

Theory- It immediately terminates the process. The process does not get a chance to save data or clean resource. It should be used only when normal kill fails.

7] Signals in linux process termination

Linux uses signals to control processes.

Important signals are:

- ① SIGTERM (15): Normal termination
- ② SIGKILL (9): Force Termination
- ③ SIGSTOP: Pause a process
- ④ SIGCONT: Continue a stopped process

The kill command send these signal to process

8] Importance of killing processes

killing unwanted process is important for:

- ① Reducing CPU load
 - ② Freeing memory
 - ③ Improving system performance
 - ④ Preventing system crashes
 - ⑤ Managing server stability
- System administrator used these command daily to keep health and responsive.